

HEALTHCONNECT EXPERIENCE LAB

Week 7 Project Summary

Testing, Refinement & End-to-End Validation — Data Analytics Track

Project Information	Details
Author	Nancy Lee YIMBERE ALAPINI
Track	Data Analytics
Programme	AnalystLab Africa — Premium Data Analytics Internship
Project	HealthConnect Experience Lab
Week 7 status	PASS Analytics ready for Week 8 integration

Purpose. This summary complements the main Week 7 Analytics Testing & Refinement Report. It captures the decisions, changes and validation outcomes required by AnalystLab without repeating the full analytical evidence.

1. Week 7 Project Summary — 16 Required Elements

#	AnalystLab Requirement	Week 7 Summary
1	Planned tests	Revalidate core KPIs; challenge the evidence hierarchy; test dashboard values, filters and interpretation; reassess recommendations; complete Analytics × Data Science cross-track validation.
2	Actual tests completed	14 Analytics tests (T01–T14) covering KPI recalculation, lead time, previous no-shows, combined prioritisation, dashboard reconciliation, filter scenarios, reminder status, distance, interaction/robustness, recommendations and control variables; plus 2 cross-track feature-ablation tests.
3	Most important results	Overall NSR = 51.15%; Reminder Coverage = 72.68%. Booking Lead Time remained the strongest standalone signal: 39.13% NSR at 0–30 days vs 63.95% at 31–60 days (+24.82pp). Recorded Previous No-Shows remained complementary: 46.30% vs 57.83% (+11.53pp).
4	Issues identified	A genuine dashboard interpretation risk was found: interactive subgroup filtering could coexist with static reference-population conclusions. The recommendation cycle also needed a clearer post-action evaluation step.
5	Refinements made	Separated MONITOR as interactive from DIAGNOSE and PRIORITIZE as validated reference views; removed context-vulnerable static elements; added explicit page guidance; strengthened recommendations to Decision → Action → Monitoring KPI → Evaluation → Extend / Adjust / Stop.
6	Retest results	Dashboard refinements passed scenario retesting. KPI and dashboard values reconciled. Recommendation logic passed reassessment. No critical unresolved Analytics inconsistency remained.
7	Tracks collaborated with	Data Science — HC-POD 01; collaborator: AYDEN NGNINTEDEM DEMANOU.
8	Collaborative test	Analytics requested controlled testing of long_lead_followup and distance_to_clinic_km. Data Science used with-vs-without ablation under the same Logistic Regression pipeline, grouped validation and significance/business-lift checks.
9	What changed	long_lead_followup was removed from the Week 8 candidate feature set because no meaningful incremental predictive value was demonstrated. distance_to_clinic_km was retained because it showed a small but consistent multivariable contribution.
10	Validation outcomes	Core KPI definitions, analytical evidence hierarchy, dashboard architecture, business insights, recommendations and Analytics/DS interpretation were validated. Dashboard remained frozen after final review.
11	Main challenges	Separating exploratory interactivity from validated reference conclusions; avoiding causal overclaiming; distinguishing statistical detectability from business relevance; reconciling weak standalone Analytics signals with multivariable predictive contribution.
12	Key decisions	Keep Booking Lead Time as PRIMARY; Recorded Previous No-Shows as SUPPORTING/COMPLEMENTARY; Reminder Status and Distance as SECONDARY/CONTEXTUAL. Do not create a Follow-up-specific lead-time rule or deterministic individual risk score.
13	Remaining limitations	Appointment-level rather than reliable longitudinal patient analysis; observational design; some missing Distance/Waiting values; sparse extreme segments; intervention effectiveness not yet established; editable-PBIX governance.
14	Remaining dependencies	No unresolved Week 7 Analytics → Data Science testing dependency remains. Week 8 must preserve the validated evidence hierarchy and interpretation guardrails during final integration.
15	Contribution to HealthConnect	Week 7 converted Week 6 outputs into a tested, traceable decision-support solution: stronger KPI assurance, safer dashboard interpretation, evidence-proportional recommendations, a simpler candidate feature set and clearer Analytics/DS integration.
16	Work remaining before Week 8 / final integration	Preserve evidence hierarchy and non-causal/non-deterministic guardrails; integrate Analytics and Data Science findings without conflation; retain Test & Measure for interventions; maintain consistency across report, dashboard, validation evidence, cross-track evidence, GitHub/Drive and final presentation.

2. Decision & Validation Snapshot

Decision Area	Final Week 7 Position	Status
Booking Lead Time	Primary standalone analytical differentiation signal.	VALIDATED
Recorded Previous No-Shows	Complementary prioritisation information; not deterministic.	VALIDATED
Reminder Status	Weak observational differentiation; no causal claim.	VALIDATED
Distance to Clinic	Secondary/contextual in Analytics; retained as a modest multivariable DS feature.	VALIDATED
Lead Time × Appointment Type	No supported distinct interaction; no Follow-up-specific rule.	VALIDATED
Dashboard architecture	MONITOR interactive; DIAGNOSE and PRIORITIZE validated reference views.	REFINED + VALIDATED
Recommendations	Prioritise → Refine → Test & Measure, with explicit evaluation cycle.	REFINED + VALIDATED
Cross-track feature set	long_lead_followup removed; distance_to_clinic_km retained.	TESTED + VALIDATED

3. Week 8 Readiness

Readiness Question	Final Answer
What is now validated?	Core Analytics KPIs, evidence hierarchy, dashboard, decision-support logic, recommendations and cross-track interpretation.
What remains uncertain?	Primarily intervention effectiveness, plus known structural and observational limitations.
What must be completed before final integration/presentation?	Evidence-consistent integration, preservation of guardrails, documentation consistency and final cross-deliverable QA.
Requires another track?	None currently open for Data Analytics from Week 7 testing.

FINAL STATUS: WEEK 7 DATA ANALYTICS PASS | READY FOR WEEK 8 INTEGRATION

4. Closing Statement

Week 7 did not replace the Week 6 analysis; it tested it. The resulting HealthConnect Analytics solution is now more defensible because its core metrics were independently revalidated, genuine dashboard interpretation risks were corrected and retested, recommendations were tied to an explicit evaluation cycle, and cross-track testing changed real modelling decisions. Remaining limitations are documented and controlled rather than hidden. The Data Analytics workstream is ready for Week 8 integration.